Problem 5: There is a N×N chessboard. It has a light on each cell. When it is turned on, a distinct cell would light up every second and go dark. The cells light up in the sequence shown in the diagram below:

	26	27			
5	25	24	23	22	21
4	10	11	12	13	20
3	9	8	7	14	19
2	2	3	6	15	18
1	1	4	5	16	17
	1	2	3	4	5

You will be given a number, which denotes the second on which, which cell would light up.

[Light 0], 1008)

Answer format: (x,y). For example:

5th second lights up (3,1).

Solution:

Lets evaluate the possible solutions.

$25 \rightarrow 5 \times 5$. So $\sqrt{25}$ gives us the value of y.

But $\rightarrow 16 \rightarrow 4 \times 4$. $\sqrt{16}$ gives us the value of x.

For $24 \rightarrow \sqrt{24} = 4.86...$ but $\lceil \sqrt{24} \rceil = 5$. again, it is correct for the value of y.

For, 25 the value of x is 1.

For, 24 its $2.23 \rightarrow 3.22 \rightarrow 4$.

So, the value of x is basically the difference from the next squared number +1.

But what about 16? Here $y = \sqrt{16} = 4$.

Also for, 15, $\lceil \sqrt{15} \rceil = 4$, again which is the value of y.

~~For y of 15, $\lceil \sqrt{16} \rceil$'s y -1 = 15's~~

For y of 16, $1 = |16 - \sqrt{16} * \sqrt{16}| + 1$

For y of 15, $2 = |15 - \lceil \sqrt{15} * \sqrt{15} | + 1$

$= |-1| + 1 = 2$

For y of 14, $3 = |14 - \lceil \sqrt{14} \rceil * \lceil \sqrt{14} \rceil| + 1 = 3$

To get positive values and to avoid ~~modulus~~ absolute we can write,

$$1 = |16 - \lceil\sqrt{16}\rceil\lceil\sqrt{16}\rceil| + 1 \quad \text{as} \quad 1 = \lceil\sqrt{16}\rceil\lceil\sqrt{16}\rceil - 16 + 1$$

We now know that the ceiling of a root of the number and the (difference from the adjacent squared number) +1 produced x and y or y and x.

In case of $\sqrt{26}$, $\lceil\sqrt{26}\rceil\lceil\sqrt{26}\rceil - 26 + 1 = 11$ and $\lceil\sqrt{26}\rceil = 6$.

here, the value $y = 6$ but 11 is not x.

In the previous cases, $\lceil\sqrt{z}\rceil \times \lceil\sqrt{z}\rceil - z + 1$ was always less than $\lceil\sqrt{z}\rceil$. Now, the fact is, the adjacent squared number is not $6 \times 6 = 36$, rather it 25.

Now, $26 - 25 = 1$, which is it's x.

$$\therefore \; 1 = 26 - (\lceil\sqrt{26}\rceil - 1)(\lceil\sqrt{26}\rceil - 1).$$

Same for, $27.\ 2 = 27 - (\lceil\sqrt{26}\rceil - 1)(\lceil\sqrt{26}\rceil - 1)$

To sum it up, $\sqrt{z}$ = even, then

$$x = \lceil\sqrt{z}\rceil \text{ and } y = \lceil\sqrt{z}\rceil\lceil\sqrt{z}\rceil - z + 1 \text{ when}$$

$$\lceil\sqrt{z}\rceil\lceil\sqrt{z}\rceil - n < \lceil\sqrt{z}\rceil \text{ but } y = z - (\lceil\sqrt{z}\rceil - 1)(\lceil\sqrt{z}\rceil - 1)$$

and when $\sqrt{z}$ = odd, then

$$x \xleftrightarrow{\text{interchange}} y$$

x, y are retrieved from same method as $\sqrt{z}$ = even but only interchanged just.

Pre-assumptions:

1. Honestly, I simply just played with the numbers. So, I guess the pre-assumption I have is that there is a lot of time to play with each individual number and if there exists any relationship among them at all. Taking time.